

Pre-read study material

City Scale Green Building & Retrofitting Analysis

1. Basic Concept of Green Building

- Sustainable Architecture - nptel.ac.in/courses/124107011 (Focus on Unit 1)

2. Green Building vs Conventional Building

- IGBC Green New Buildings overview - [IGBC | IGBC Green New Buildings Rating System](#)

3. GRIHA Rating System

- GRIHA official rating page [GRIHA Rating | GRIHA](#) and GRIHA v2019 manual <https://www.bing.com/ck/a?!&&p=3422fd8255d510ade8bac20dc76781462dc7ba53f3c5e9e6dc90be51babbee8fjmltdHM9MTc3ODYzMDQwMA&ptn=3&ver=2&hsh=4&fclid=327837a2-8239-6097-371c-21f9838b6154&psq=GRIHA+V.2019+A+GRIHA+Council+Publication+Volume&u=a1aHR0cHM6Ly93d3cuZ3JpaGFpbmRpYS5vcmcvc2l0ZXMvZGVmYXVsdC9maWxlcY8yMDI1LTEwL0dSSUhBX1ZvbHVtZV8xX0luc2lkZV9QYWdlc18xLnBkZg>
- *The student should study only the key criteria related to: Sustainable site planning, Building design optimization, Energy performance, Renewable energy integration, Water conservation, Wastewater treatment and reuse, Sustainable building materials, Indoor environmental quality, Occupant thermal and visual comfort*

4. ECBC 2017 – Energy Conservation Building Code

- <https://www.bing.com/ck/a?!&&p=05380c96b577e0432c90727b05b79d8d7c531c74c50c7e62b0a80c98143cffa1jmltdHM9MTc3ODcxNjgwMA&ptn=3&ver=2&hsh=4&fclid=327837a2-8239-6097-371c-21f9838b6154&psq=ecbc+2017&u=a1aHR0cHM6Ly9oYXJlZGEuZ292LmluL2RvY3VtZW50LzltZWNiYy1ib29rLWZpbmFsLTlwMTcv>
- *The student should understand the minimum energy efficiency requirements for commercial buildings in India and should study only the following chapters: (Chapter 4: Building Envelope, Chapter 5: HVAC, Chapter 6: Lighting, Chapter 7: Electrical Power)*

5. Energy Conservation and Sustainable Building Code 2024

- <https://www.beeindia.gov.in/WriteReadData/L45218/7534511430467336.pdf>
- *The student should understand the latest direction of Indian building energy and sustainability compliance and focus on: Energy efficiency, Sustainable site, Water efficiency, Material efficiency, Indoor environmental quality, Operation and maintenance.*

6. NPTEL Lectures

- Sustainable Architecture, IIT Roorkee nptel.ac.in/courses/124107011

- *Targeted topics: Introduction to sustainable architecture, Climate responsive design, Passive design, Daylighting, Green building rating systems (The student should watch selected lectures only).*

Sustainable Materials and Green Buildings, IIT Delhi nptel.ac.in/courses/105102195

- *Targeted topics: Sustainable construction materials, Embodied energy, Life-cycle thinking, Recycled materials, Low-impact materials, Material selection criteria (The student should understand sustainable material selection and environmental impact).*

7. Review Paper

- **A review on insulation materials for energy conservation in buildings**

<https://doi.org/10.1016/j.rser.2017.02.034>

8. Case Study

- **IGBC or GRIHA-rated Indian building case study**

- <https://www.bing.com/ck/a?!&&p=f4579dce380f645aa1e6b55404e1fba1224b6dd851baa7755171aadbac54a9c6JmltdHM9MTc3ODcxNjgwMA&ptn=3&ver=2&hsh=4&fclid=327837a2-8239-6097-371c-21f9838b6154&psq=iitgn+certified+by+GRIHA+HOW+AND+WHICH+PARAMETER+FOCUS&u=a1aHR0cHM6Ly93d3cucmUtdGhpbmtpbmd0aGVmdXR1cmUuY29tL2Nhc2Utc3R1ZGllcy9hMjkxNC1paXQtZ2FuZGhpbmFnYXItaW5kaWFzLWZpcnN0LWNhbXB1cy10by1yZWNIaXZlLWEtZml2ZS1zdGFyLWdyaWhhLWxkLXJhdGluZy8>